

BOURBOT

Background & Motivation

- Whiskey barrels at large distilleries are aged in **rickhouses**, where barrels are **transported by hand** to their aging rack.



A rickhouse worker manually rolls a barrel [1]

- Nearly **1000 reported injuries** from 2010-2019.



The bung protrudes slightly from the barrel [1]

- Barrels must be precisely **clocked** so the **bung** faces upward to avoid leaks.



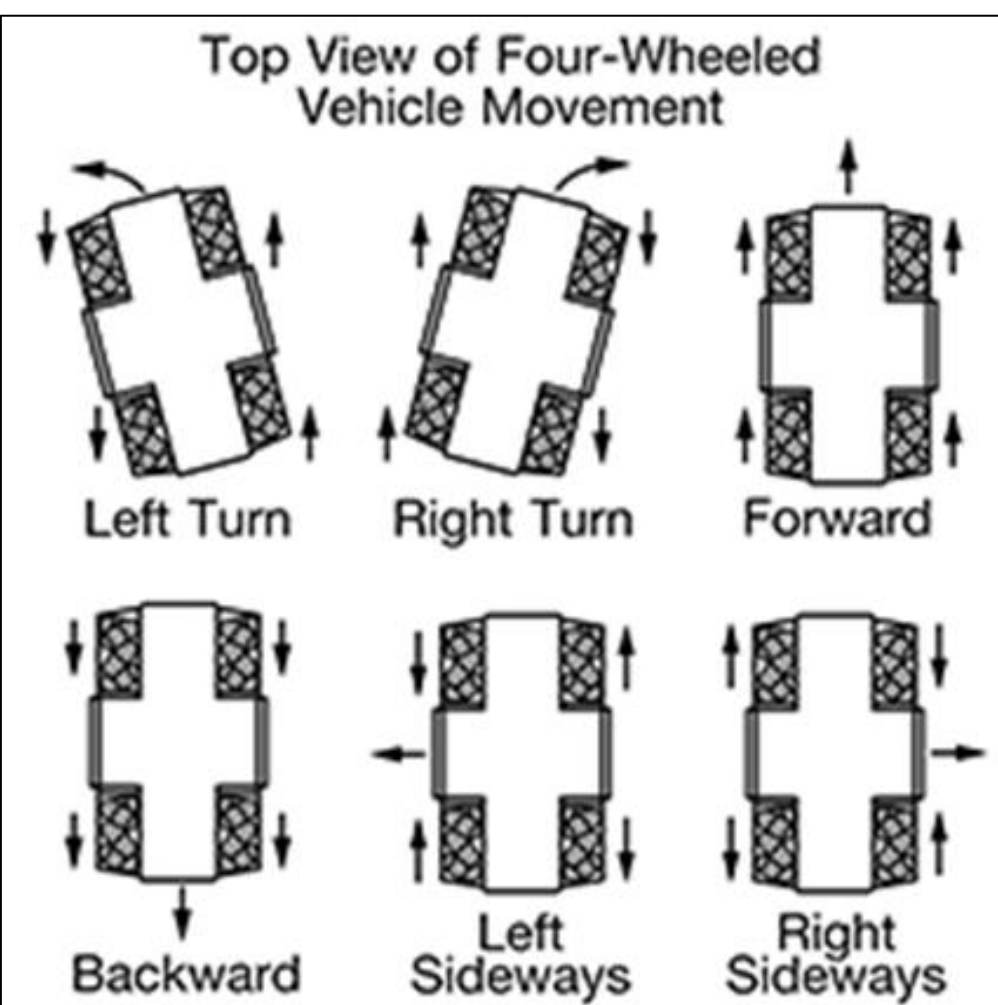
A rickhouse hallway [2]

- Current technology is insufficient due to **space constrictions** and clashes with **traditional methods**.

Problem Statement

To design a semi-automated wheeled robot that improves the **transportation** and **clocking** of 550-pound whiskey barrels within rickhouses - increasing **efficiency** and **safety** while maintaining the **traditions** of the whiskey-making process.

Design Overview / Challenges



Mecanum Wheel Directional Control [3]

Motion: Motorized mecanum wheels roll the barrel on the ground while allowing omni-directional motion while approaching and releasing the barrel.

Clocking: The ball screw lifting mechanism and active roller interface lifts the barrel and rotates it in midair.



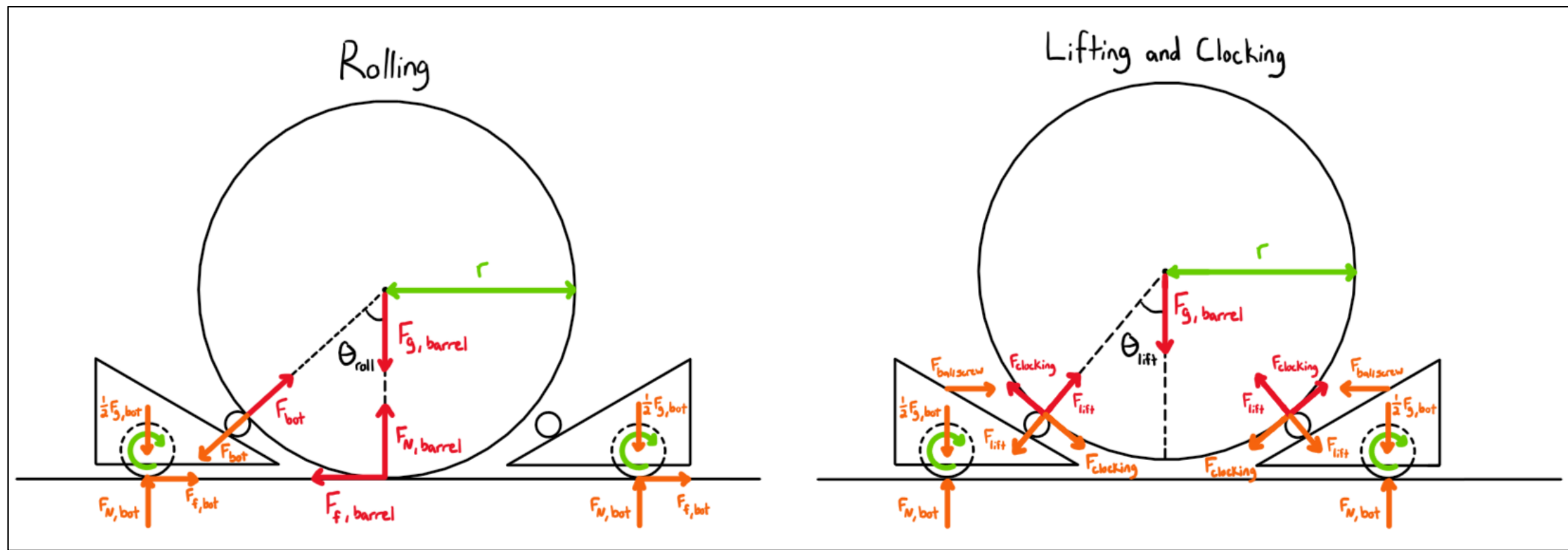
Barrel Clocking Motion with Rollers Highlighted

- Challenges:**
- Lifting the barrel while minimizing chassis deformation
 - Maintaining stability and avoiding bung interference while rolling and clocking

Motor Torque Analysis

ASSUMPTIONS AND BOUNDARY CONDITIONS:

- Full Barrel = Solid Cylinder
 - No effects of sloshing
- Even distribution of weight on wheels of Bourbot
- Flat surface & 1° incline
- Factor of Safety = 1.75
- Pushing from rest



Free Body Diagrams

Iterative Testing of Design Equations



- Total torque needed by each active roller ($\Sigma F_{friction + inertia}$)
 - Total torque needed by each wheel to push ($\Sigma F_{friction + inertia}$)
- Determine total tension required to lift barrel (ΣF_x)

Resulting Mechanical Requirements

FINAL VALUES

- $W_{total} = 175\text{lbs}$
- $\theta_{contact} = 45^\circ$
- $a_{bourbot} = 0.1\text{ ft/s}^2$
- $v_{bourbot} = 3.94 - 4.59\text{ ft/s}$
- $\tau_{max\text{ per wheel}} \sim 5.35\text{ lb-ft}$
- $\tau_{push} > \tau_{push\text{ while lifting}}$
- $F_{lifting} = 284.65\text{ lbf}$

Ball Screw Torque Required for Lifting

Variables:

- τ = ball screw torque
- F_a = axial force = ΣF_x
- l = screw lead
- η = ball screw efficiency (~80%)



Ball Screw Torque

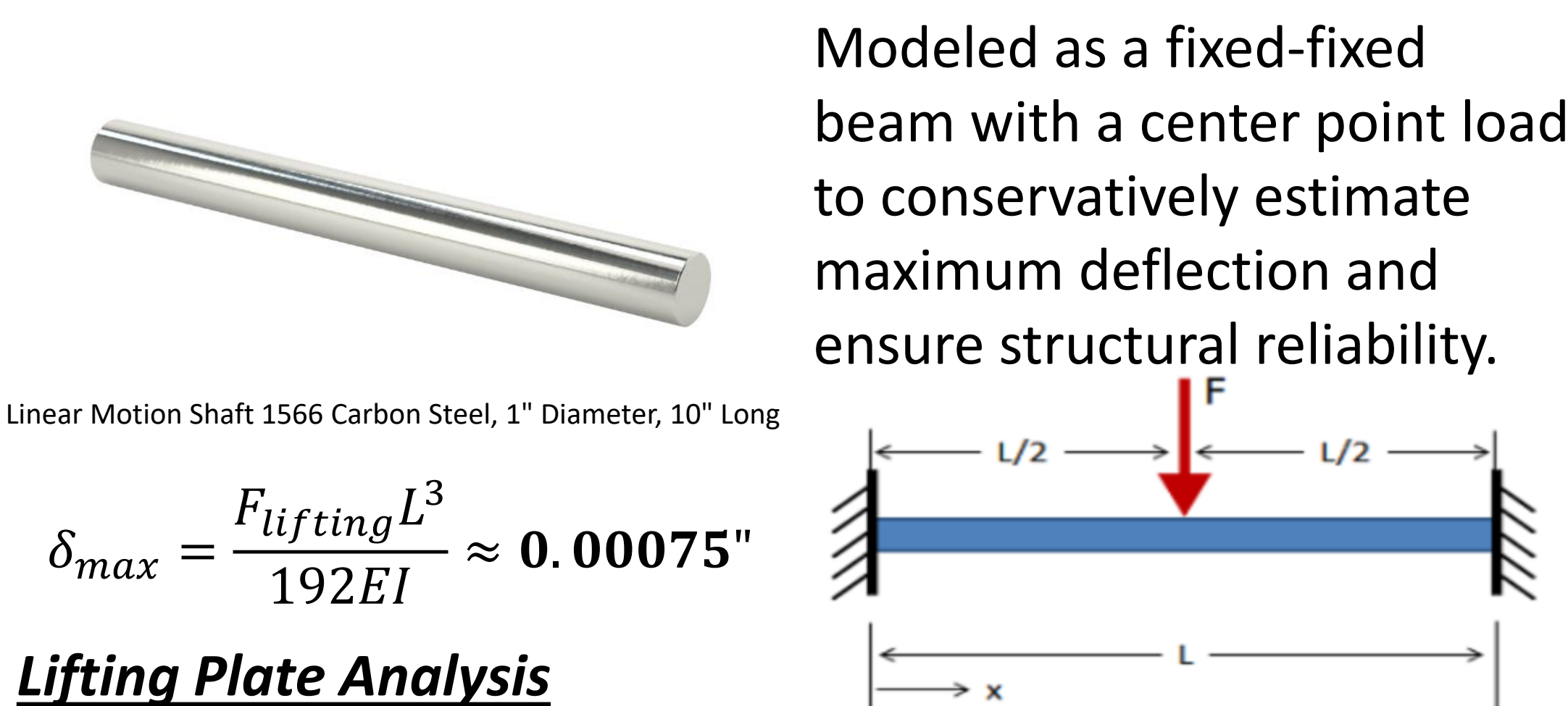
$$\tau = \frac{F_a l}{2\pi\eta}$$

Using total tension to lift barrel (ΣF_x) and FOS = 3: $\tau_{motor} \approx 2.79\text{ lb-ft}$

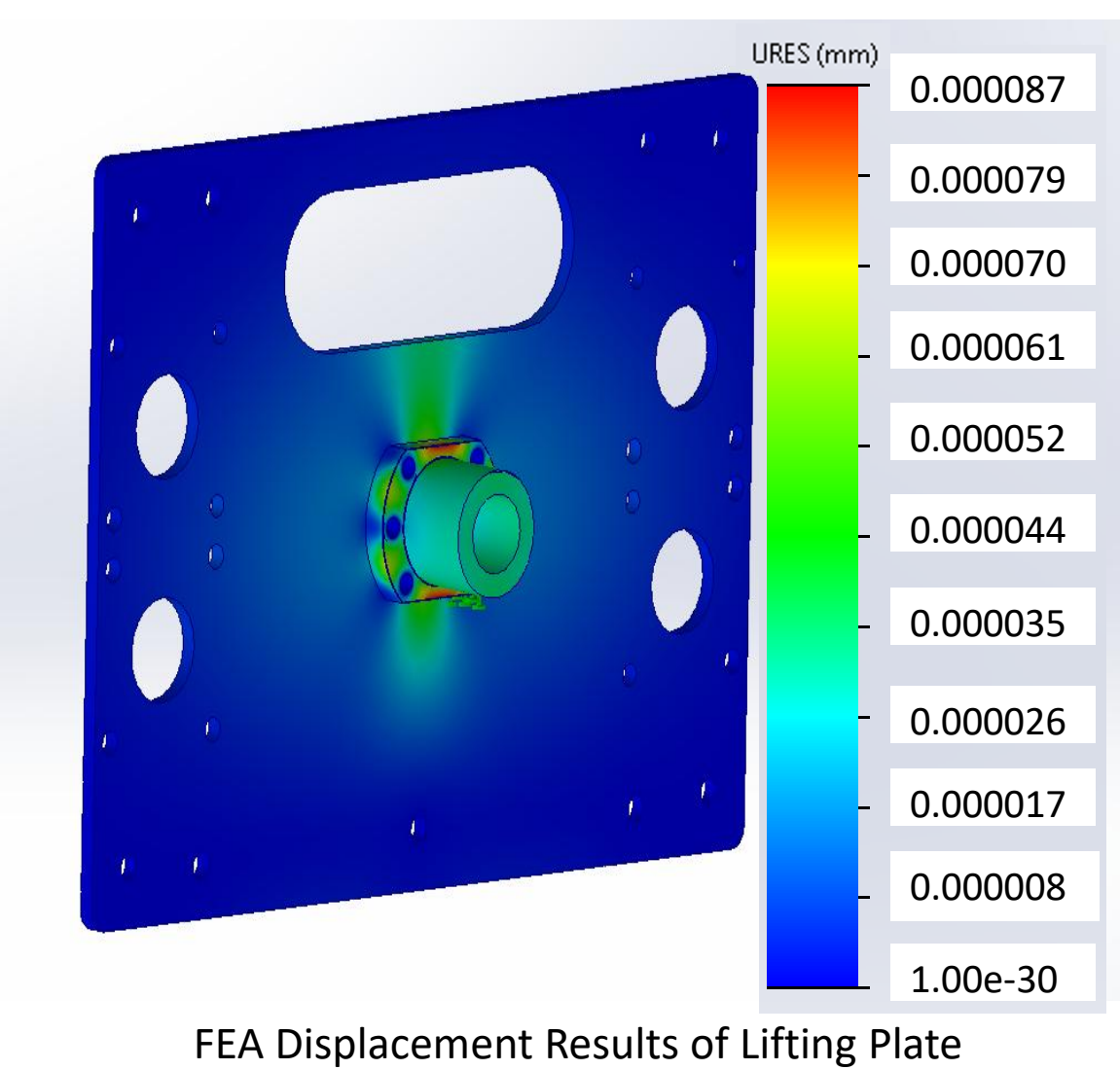
Lifting Analysis

Shaft Deflection Analysis

To maintain chassis alignment during lifting, we evaluated three main criteria: **required extension length, shaft load forces from the lifted barrel, and beam material selection.**



Lifting Plate Analysis



Fixed-fixed beam under center point load

Analytical Model:
Modeled as a simply supported beam under center load
Deflection: 7.0×10^{-6} in

FEA Validation:
Realistic conditions in SOLIDWORKS Simulation
Deflection: 3.45×10^{-6} in

Acknowledgements

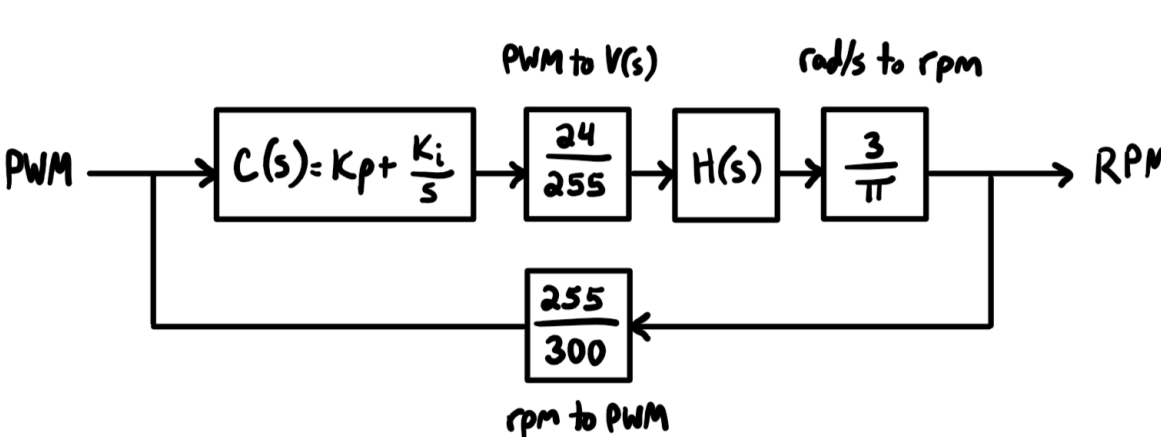
The Bourbot team thanks Dr. Andrew Gouldstone, Dr. Mehdi Abedi, Dr. Scott Julien, the Capstone Lab staff, GrandTen Distilling, and Bully Boy Distillers for their valuable insight, support and resources on this project.

PID Control

Assumptions: Rigid shafts, $J_{encoder}$ is negligible, $B_{bearing}$ is negligible, direct power transfer between sprockets, $k_t = k_e$

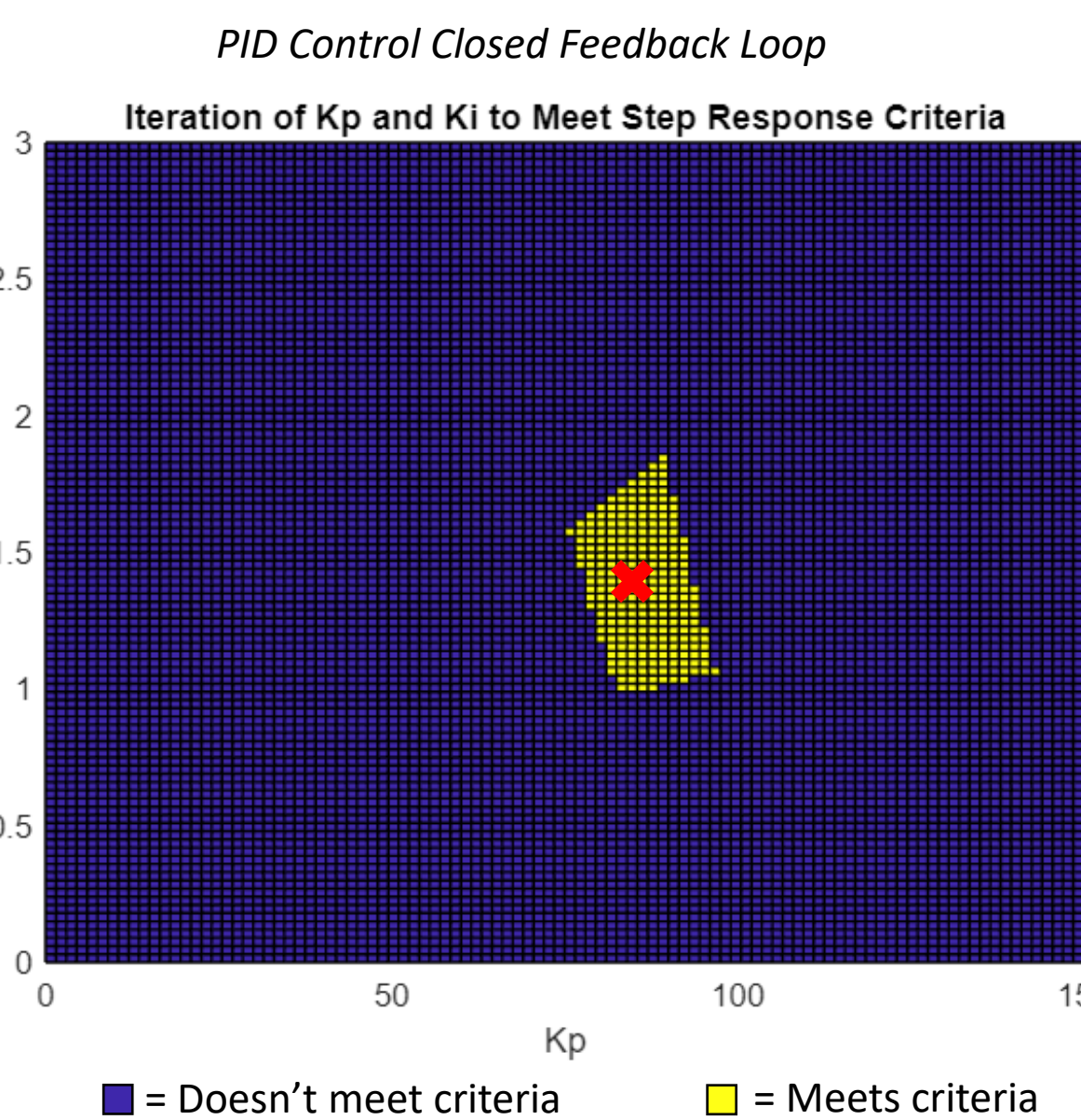
DC Motor Transfer Function:

$$\frac{\dot{\theta}(s)}{V(s)} = \frac{k}{(Js + B)(Ls + R) + k^2}$$



For the Bourbot:

$$H(s) = \frac{0.2089}{s + 0.02}$$



Design Criteria:

- Stable
- $2.75s < T_{settle} < 3.25s$
- No Overshoot
- $99.5\% \leq \text{Peak} \leq 100\%$

Result:

Kp = 85 and Ki = 1.5 meet the given criteria

References

- [1] "Could You Be a Ricker Rider? — THE BITTER SOUTHERNER." Accessed: Apr. 12, 2025. [Online]. Available: <https://bittersoutherner.com/could-you-be-a-ricker-rider-kentucky-bourbon-whiskey>
- [2] "Dancing Goat Rickhouse." Ideal Builders. Accessed: Apr. 12, 2025. [Online]. Available: <https://idealbuilders.com/work/dancing-goat-rickhouse>
- [3] "401e-four-wheeled-vehicle-movement-digital-master1571175304-g98@1x_63708768823417452.png (498x499)." Accessed: Apr. 01, 2025. [Online]. Available: <https://www.mcmaster.com/mvc/Contents/gv/ImageCache/e01/e01e-four-wheeled-vehicle-movement-digital-master1571175304>

Design Validation

To confirm the Bourbot's ability to lift the empty barrel, we compared specs with the minimum requirements and tested functionality of the mecanum wheels.



Bourbot lifting empty barrel

Methods:

- Lifting Test: The Bourbot lifted an empty barrel.
- Driving Test: We evaluated the Bourbot's directional movement.

Test Conditions: Testing was performed under controlled conditions and flat flooring to ensure repeatability.

Findings: The Bourbot lifted the **120lb** barrel successfully and displayed varied directional movements.

Conclusion: The Bourbot demonstrates sufficient torque for lifting and delivers directional control under tight spaces.

Additional Finding: The barrel remains stable once lifted. The stepper motors for the clocking mechanism and the front right wheel require debugging in the movement code.



Spacing between the lifted barrel and the ground

Future Work / Exit Strategy

Next Steps

- Reach out to distilleries for feedback
- Create a list of suggested design changes to support future work, such as
 - Altering the roller locations
 - Strengthening the lifting shafts
 - Increasing the chassis clearance
 - Further supporting the arm connections
- Finalize prototype documentation
- Disassemble prototype and donate components to the Capstone Lab